

Semantics and Equational Axiomatisation of Quantum Communication

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1 Overview of the Presentation

What. We study the semantics of quantum programs extended with classically controlled quantum communication: sending and receiving qubits, possibly depending on a classical measurement outcome. For example, the following is naturally expressed in imperative syntax, but not in the standard quantum circuit model.

```
p: qbit = in(); q: qbit = new(|0>);
if (measure(p) == |1>) { out(q); }
```

The goal is to develop an axiomatisation of program equivalence between communicating quantum programs.

Why. The broader research programme is to develop operationally and denotationally sound verification tools for distributed quantum programming [1]. Prior work focuses on quantum process calculi (QPCs) [2, 3, 6, 12], which give a natural bisimulation-based verification technique for quantum protocols. QPCs are expressive (parallelism + communication + quantum effects) but complex. They are presented operationally lack a clear denotational reading compatible with quantum theory. Our strategy is to isolate quantum communication from concurrency, recovering a denotational reading alongside the operational one. Verification then happens through an equational axiomatisation.

How. Concretely, we model communication operations as algebraic effects, following the tradition of Plotkin and Power [8], as summarised by fig. 1. We propose a new algebraic theory for quantum communication building on [10] and study two models (operational and denotational). Finally we show the operational and denotational equalities coincide.

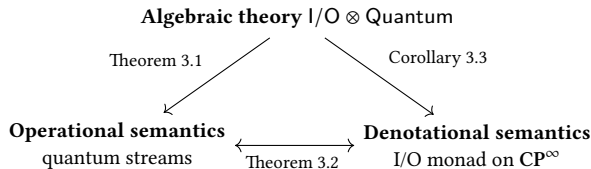


Figure 1. Three semantics of quantum communication.

*This presentation is based on joint work with Sam Staton (Oxford) [11], presented for the first time at PPlanQC 2025.

2 Background: Axiomatising Quantum Computing

We already know how to build the same triangle for qubit quantum computing without communication. Our starting point is [10]’s axiomatisation for qubit quantum computing, Quantum, based on *parameterised* algebraic theories (PAT). PATs are a generic framework for axiomatising the behaviour of computations depending on certain resources in terms of operations and equations. Here, resources will correspond to qubits.

Operations $\text{op} : (p \mid m_1 \dots m_k)$ represent computations which take p qubits, classically choose one of the k branches, say, i , and return m_i qubits. Quantum has three operations: state preparation $\text{new} : (0 \mid 1)$, unitary evolution $\text{apply}_U : (n \mid n)$ for each $2^n \times 2^n$ unitary U , and computational basis measurement & classical control $\text{measure} : (1 \mid 0, 0)$.

Terms generated from the operations follow the same intuition: a term $x_1 : m_1 \dots x_k : m_k \mid q_1 \dots q_p \vdash c$ takes p input qubits $q_1 \dots q_p$ and calls one of the k continuations $x_1 \dots x_k$ with the corresponding number of qubits. We can thus write for example the program

$$x : 1, y : 1 \mid q_1, q_2 \vdash \text{measure}(q_1, x(q_2), y(q_2))$$

which measures q_1 and continues with x or y with q_2 based on the measurement outcome.

Equations are given in the form $\Gamma \mid \Delta \vdash c = c'$. Quantum for example specifies the effect of measuring a $|0\rangle$ qubit using the following equation:

$$x : 0, y : 0 \mid \cdot \vdash \text{new}(q.\text{measure}(q, x, y)) = x$$

Models: Denotational and Operational [10] proceeds to define a denotational model for the Quantum theory, where each term $x_1 : m_1 \dots x_k : m_k \mid q_1 \dots q_p \vdash c$ is interpreted as a complete positive unital (CPU) map

$$\llbracket c \rrbracket : \bigoplus_{i=1}^k \mathcal{M}_{2^{m_i}}(\mathbb{C}) \rightarrow \mathcal{M}_{2^p}(\mathbb{C}).$$

Remarkably, this model is *fully complete*: every CPU map of this type is the interpretation of some term c , and two CPU maps are equal iff their term representations are derivably equal in the equational theory.

We can also give an equivalent operational model for Quantum. Briefly, configurations are of the form $\langle \rho, c \rangle$ where ρ is the program state, c the running program, and transitions $\rightarrow_p$ are probabilistic. We can then formulate in terms of $\rightarrow_p$ an operational notion of program equivalence $\simeq_{qu}$ equivalent to denotational equality.

3 Semantics for Quantum Communication

We proceed to extend our language with communication primitives, and strive to recover the same triangle.

3.1 An Algebraic Theory for Quantum Communication

Extending Quantum with two operations: $\text{in} : (0 \mid 1)$ and $\text{out} : (1 \mid 0)$ lets us represent classically controlled communication. We can express the example program as $x : 0 \mid \cdot \vdash \text{in}(p.\text{new}(q.\text{measure}(p, x, \text{out}(q, x))))$. We further add equations to make in and out commute with every Quantum operation. What we now obtain is $\text{I/O} \otimes \text{Quantum}$, where $\otimes$ denotes the commutative combination of two theories. This choice is motivated by quantum theory. For instance,

$$\text{measure}(a, \text{out}(b, x), \text{out}(b, y)) = \text{out}(b, \text{measure}(a, x, y))$$

is required to derive the equivalence between: preparing a Bell pair, sending one qubit and discarding the other; and tossing a fair coin and sending the outcome as a qubit.

3.2 An Operational Model

Concretely, the notion of communication that our theory soundly axiomatises can be demonstrated through an operational model. We work in CP , the category of finite dimensional C^* algebras and complete positive maps, constructed as in [9].

The operational semantics is based on a *quantum stream*, specifying the environment our program communicates with. The idea is to have a global quantum state $\rho : \mathbb{C} \rightarrow S \otimes \text{qbit} \otimes H$, where S is the program state, H is the hidden stream state, and qbit holds the next input qubit. Then, a stream is a state machine reacting to the program's communications: formally, a stream t is specified by a hidden state type H_σ and transition maps $T_{\text{in}}(\sigma), T_{\text{out}}(\sigma)$ for every communication trace $\sigma \in \{\text{in}, \text{out}\}^*$.

We can then build an operational semantics with configuration $\langle \rho, \sigma, c \rangle$, where ρ is the global state, $\sigma \in \{\text{in}, \text{out}\}^*$ is the communication trace, and c is the program. Given a stream specification t , we can give a small-step transition relation $\rightarrow_p^t$ defined inductively on the program. For example, for the output operation, the rule is as in fig. 2: the output is performed, and the stream state is updated according to t . Finally, $\simeq_{qu}$ extends naturally to communicating quantum programs and gives a sound model of $\text{I/O} \otimes \text{Quantum}$.

Theorem 3.1. $\Gamma \mid \Delta \vdash c = c' \implies \Gamma \mid \Delta \vdash c \simeq_{qu} c'$.

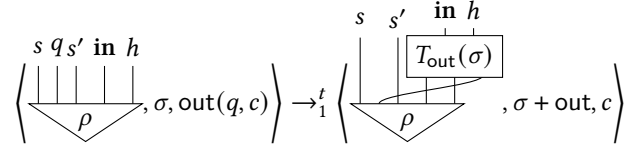


Figure 2. Operational rule for out

3.3 A Denotational Model

The Quantum I/O Monad In the tradition of algebraic effects [4, 8], we build an I/O monad $(T(X) = \mu Y. [\text{qbit}^{\text{in}}, Y] \oplus \text{qbit}^{\text{out}} \otimes Y \oplus X)$, where the in and out superscripts are book-keeping) directly on top of CP . On the one hand, we can simplify the definition of T by unrolling the fixed point and by noting that $[\text{qbit}^{\text{in}}, Y] \cong \text{qbit}^{\text{in}} \otimes Y$ by compact closure and self-duality:

$$T(X) = \bigoplus_{\sigma \in \{\text{in}, \text{out}\}^*} (\text{qbit}^\sigma \otimes X)$$

where $\text{qbit}^{s_1 \dots s_n} = \text{qbit}^{s_1} \otimes \dots \otimes \text{qbit}^{s_n}$, $s_i \in \{\text{in}, \text{out}\}$. On the other hand, T is not well-defined because CP only has finite biproducts. To make it well-defined, we follow [7] and complete CP with infinite biproducts using a dcpo completion, obtaining CP^∞ . Now T is well-defined as a functor $\text{CP}^\infty \rightarrow \text{CP}^\infty$, which is in fact a monad.

Denotational Semantics We simply interpret each $\Gamma \mid \Delta \vdash c$ term as a morphism $\llbracket c \rrbracket : \llbracket \Delta \rrbracket \rightarrow T(\llbracket \Gamma \rrbracket)$ in the Kleisli category of T . The interesting case is the input operator $\llbracket \text{in} \rrbracket : \mathbb{C} \rightarrow T(\text{qbit})$, defined as

$$\llbracket \text{in} \rrbracket \triangleq \text{inj}^{\text{in}} \circ \left[\text{qbit}^{\text{in}} \bigcup \text{qbit} \right]$$

where the ‘cup’ is the unnormalised Bell state. Intuitively, $\llbracket \text{in} \rrbracket$ is supposed to take a new input; the cup bends the input wire up so it appears on the output side, as specified by the monad. Remarkably, this denotation corresponds exactly to the operational notion of equality, $\simeq_{qu}$. We thus get:

Theorem 3.2. $\Gamma \mid \Delta \vdash c \simeq_{qu} c' \iff \llbracket c \rrbracket = \llbracket c' \rrbracket$.

Corollary 3.3. $\Gamma \mid \Delta \vdash c = c' \implies \llbracket c \rrbracket = \llbracket c' \rrbracket$.

4 Limitations and Future Work

We have obtained a sound axiomatisation $\text{I/O} \otimes \text{Quantum}$ for one notion of quantum communication, presented in two equivalent ways: operationally and denotationally – a first step towards equational verification of quantum protocols. However, unlike the quantum computing case, we do not have completeness – whether $\llbracket c \rrbracket = \llbracket c' \rrbracket \implies \Gamma \mid \Delta \vdash c = c'$ is open. Achieving this is our immediate next step. Beyond that, we are interested in connecting this work back to quantum process calculi (e.g. [2]) and higher-order quantum computing [5, 7].

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